

FRANCISCA VASCONCELOS

francisca@berkeley.edu

<https://franciscavasconcelos.github.io/>

<https://www.linkedin.com/in/franciscavasconcelos>

EDUCATION

University of California Berkeley 2022-Present

Degree PhD Student **Advisors:** Prof. Michael I. Jordan and Prof. Umesh Vazirani **GPA:** 4.0/4.0
Fellowships National Science Foundation Graduate Research Fellowship (Fall 2022 - Spring 2025)
 Paul & Daisy Soros Fellowship for New Americans (Fall 2024 - Spring 2026)

University of Oxford – Rhodes Scholar 2021-2022

Degree Master of Studies (MSt) in Philosophy of Physics **College:** Keble **Mark:** Pass
Tutors Prof. Owen Maroney, Prof. James Reed, Prof. Oliver Pooley

University of Oxford – Rhodes Scholar 2020-2021

Degree Master of Science (MSc) in Statistical Science **College:** Keble **Mark:** Merit
Thesis *Uncertainty in Implicit Neural Representations for Medical Imaging* **Supervisor:** Prof. Yee Whye Teh

Massachusetts Institute of Technology Class of 2020

Degree BS in Electrical Engineering & Computer Science, Physics **Humanities Concentration:** Philosophy
Honors Phi Beta Kappa ♦ Tau Beta Pi ♦ Sigma Xi **Societies:** IEEE ♦ APS ♦ SWE **GPA:** 4.9/5.0
Thesis *Extending Quantum State Tomography for Superconducting Q. Processors* **Supervisor:** Prof. William Oliver

Torrey Pines High School Class of 2016

PUBLICATIONS

Publications sorted by category and ordered in reverse chronological order, with relevant presentations and awards listed.
 * denotes equal contribution or alphabetical ordering. (Note that TCS papers are typically alphabetical order.)

- Journal** ♦ **“A Quadratic Speedup in Finding Nash Equilibria of Quantum Zero-Sum Games.”**
 Francisca Vasconcelos*, Emmanouil-Vasileios Vlatakis-Gkaragkounis*, Panayotis Mertikopoulos,
 Georgios Piliouras, and Michael I. Jordan.
Quantum, April 2025. [[arXiv:2311.10859](#)]
 ♦ **Long Talk** (~6% acceptance) at *Quantum Techniques in Machine Learning (QTML) Conference 2023*
 (CERN, Switzerland) [[Slides](#)]
 ♦ I also presented at: Surfing the Ocean ERC Seminar [[Video](#)]
- ♦ **“UncertalNR: Uncertainty Quantification of End-to-End Implicit Neural Representations for Computed Tomography.”**
 Francisca Vasconcelos*, Bobby He*, Nalini Singh, and Yee Whye Teh.
Transactions on Machine Learning Research (TMLR), April 2023. [[arXiv:2202.10847](#)] [[Talk](#)] [[Slides](#)] [[Code](#)]
 ♦ **Oral** (~6% acceptance) at *MedNeurIPS 2021 Workshop* (Virtual) [[Video](#)]
 ♦ **Poster** at *NeurIPS 2021 Bayesian Deep Learning Workshop* (Virtual) [[Poster](#)]
- ♦ **“Generating Spatially Entangled Itinerant Photons with Waveguide Quantum Electrodynamics.”**
 Bharath Kannan, Daniel Campbell, Francisca Vasconcelos, Roni Winik, David Kim, Morten
 Kjaergaard, Philip Krantz, Alexander Melville, Bethany Niedzielski, Jonilyn Yoder, Terry Orlando,
 Simon Gustavsson, and William Oliver.
Science Advances, Vol. 6, No. 41, 2020. [[arXiv:2003.07300](#)] [[MIT News](#)]

- ◇ **“Impact of ionizing radiation on superconducting qubit coherence.”**
Antti Vepsäläinen, Amir Karamlou, John Orrell, Aksunna Dogra, Ben Loer, John Orrell, [Francisca Vasconcelos](#), David Kim, Alexander Melville, Bethany Niedzielsk, Jonilyn Yoder, Simon Gustavsson, Joseph Formaggio, Brent VanDevender, and William Oliver.
Nature, Vol. 584, pages 551–556. 2020. [[arXiv:2001.09190](#)] [[MIT News](#)] ◇

Paper

- ◇ **“Improved Lower Bounds for QAC⁰.”**
Malvika Raj Joshi*, Avishay Tal*, [Francisca Vasconcelos](#)*, John Wright*
To appear in the proceedings of the *58th Annual ACM Symposium on Theory of Computing (STOC)*, 2026. [[arXiv:2512.14643](#)]
- ◇ **“Random Unitaries in Constant (Quantum) Time.”**
Ben Foxman*, Natalie Parham*, [Francisca Vasconcelos](#)*, Henry Yuen*
In the proceedings of the *17th Annual Innovations in Theoretical Computer Science (ITCS)*, 2026.
- ◇ **Talk** at the *Quantum Information Processing (QIP) Conference 2026* (Riga, Latvia).
- ◇ **“Learning shallow quantum circuits with many-qubit gates.”**
[Francisca Vasconcelos](#), Hsin-Yuan Huang
In the proceedings of the *38th Annual Conference on Learning Theory (COLT)*, 2025. [[Slides](#)]
- ◇ **Talk** at the *Quantum Techniques in Machine Learning (QTML) Conference. 2024* (Melbourne, Australia)
- ◇ I also presented at: NYU CQP, Cambridge Quantum Computing Seminar.
- ◇ **“On the Pauli Spectrum of QAC⁰.”**
Shivam Nadimpall*, Natalie Parham*, [Francisca Vasconcelos](#)*, Henry Yuen*.
In the proceedings of the *56th Annual ACM Symposium on Theory of Computing (STOC)*, 2024.
- ◇ **Talk** at the *Quantum Information Processing (QIP) Conference 2024* (Taipei, Taiwan).
- ◇ I presented at: Simons Quantum Pod, Berkeley TCS Seminar, QuSoft Seminar. [[Talk](#)] [[Slides](#)]
- ◇ **“Person-following UAVs.”**
[Francisca Vasconcelos](#) and Nuno Vasconcelos.
In the proceedings of the *2016 IEEE Winter Conference on Applications of Computer Vision*.

Article

- ◇ **“Why Quantum Education?”**
[Francisca Vasconcelos](#).
SPIE Photonics Focus, Vol. 4, pages 6-7. May/June 2023.
- ◇ **“Quantum Computing @ MIT: The Past, Present, and Future of the Second Revolution in Computing.”**
[Francisca Vasconcelos](#).
Abridged version in *MIT Undergraduate Research Journal*, Vol. 38, pages 13-24. 2020.
Full version on [[arXiv: 2002.05559](#)].

Poster

- ◇ **“Extending Quantum State Tomography for Superconducting Quantum Processors.”**
[Francisca Vasconcelos](#), Morten Kjaergaard, Terry Orlando, Simon Gustavsson, and William Oliver.
In the proceedings of the *2019 MIT MTL Microsystems Annual Research Conference*.
- ◇ Received a Top-10 Presentation Award and Best Student Presentation Award.

Preprint

- ◇ **“Constant-Depth Unitary Preparation of Dicke States.”**
Malvika Raj Joshi*, [Francisca Vasconcelos](#)*
[[arXiv:2601.10693](#)]
- ◇ **“Methods for Reducing Ancilla-Overhead in Block Encodings.”**
[Francisca Vasconcelos](#), Andras Gilyén
[[arXiv:2311.10859](#)]
- ◇ **Talk** at the *Quantum Techniques in Machine Learning (QTML) Conference. 2024* (Melbourne, Australia)
- ◇ **Talk** at the *Quantum Simulation (QSim) Conference. 2025* (New York City) [[Slides](#)]

RESEARCH TALKS

- 2026** ◇ **“Interesting Directions and Open Questions in Shallow-Depth Quantum Circuit Complexity”**
Invited talk (1 hour) at 2nd Workshop on Random Quantum Circuits at the University of Köln on August 19th.
- ◇ **“How Powerful are Shallow Quantum Circuits?”**
Invited talk (1 hour) at the Amazon Center for Quantum Computing Algorithms Weekly Meeting (hosted by Alex Dalzell) on June 9th.
- ◇ **“How Powerful are Shallow Quantum Circuits?”**
Invited talk (1 hour) at the Caltech Quantum TCS Tea Seminar (hosted by Urmila Mahadev) on May 19th.
- ◇ **“QAC⁰ and the Quest to Compute Parity”**
Invited talk (1 hour) at the Columbia Quantum Theory Group Meeting (hosted by Henry Yuen) on April 28th.
- ◇ **“QAC⁰ and the Quest to Compute Parity”**
Invited talk (1 hour) at the Yale Theoretical Computer Science Student Seminar on April 17th.
- ◇ **“QAC⁰ and the Quest to Compute Parity”**
Invited talk (1 hour) at the MIT Quantum Theory Group Meeting (hosted by Anand Natarajan) on April 8th.
- ◇ **“How Powerful are Shallow Quantum Circuits?”**
Invited talk (1 hour) at the MIT Engineering Quantum Systems Group (hosted by William Oliver) on April 7th.
- ◇ **“Constant-Depth Unitary Preparation of Dicke States”**
Invited talk (30 min) at the UC Berkeley Challenge Institute for Quantum Computing (CIQC) Seminar on April 3rd.
- ◇ **“Constant-Depth Unitary Preparation of Dicke States”**
Invited talk (1 hour) at the Simons Institute Quantum Pod Seminar (hosted by Umesh Vazirani) at UC Berkeley on March 6th.
- ◇ **“How Powerful are Shallow Quantum Circuits?”**
Invited talk (45 min) with Malvika Joshi at the UC Berkeley Theory Lunch Seminar (hosted by Avishay Tal) on February 25th.
- ◇ **“Understanding Entanglement Generation in Constant-Depth Quantum Circuits with Global Interactions”**
Invited talk (1 hour) at TU Delft QuTech (hosted by Francisco Machado) on February 4th.
- ◇ **“QAC⁰ and the Quest to Compute Parity”**
Invited talk (1 hour) at the CWI QuSoft QuLunch Seminar (hosted by Maris Ozols) on February 3rd.
- 2025** ◇ **“QAC⁰ and the Quest to Compute Parity”**
Invited talk (1 hour) at the First Quantum Cambridge-Oxford-Warwick Colloquium (QCOW) on Quantum Low-Depth Complexity at Oxford University on December 12th.
- ◇ **“Methods for Reducing Ancilla-Overhead in Block Encodings”**
Invited talk (45 min) at the University of Warwick Quantum Computing Seminar (hosted by Matthias Caro) on December 9th.
- ◇ **“Deep Thoughts on (Learning) Shallow Quantum Circuits”**
Invited talk (1 hour) at the 1st AIMS Workshop on Quantum Learning Theory in Cape Town, South Africa on October 29th.
- ◇ **“From Classical to Quantum Circuit Complexity: The Tale of QAC⁰”** [[Video](#)]
Invited talk (1 hour) at the Institute for Advanced Studies Computer Science & Discrete Math Seminar (hosted by Avi Wigderson) in Princeton on October 20th.

- ◇ **“Methods for Reducing Ancilla-Overhead in Block Encodings”**
Contributed Talk (20 min) at the Quantum Simulation (QSim) conference in NYC on August 8th.
- ◇ **“Learning shallow quantum circuits with many-qubit gates”** [[Video](#)]
Contributed Talk (12 min) at the Conference on Learning Theory (COLT) in Lyon, France on July 4th.
- ◇ **“Rethinking Quantum Advantage via the Quantum Singular Value Transform”**
Contributed Talk (25 min) at the 22nd European Conference on Foundations of Physics in Gdansk, Poland on July 1st.
- ◇ **“Learning shallow quantum circuits with many-qubit gates”**
Invited talk (1 hour chalkboard) at Cambridge University Quantum Computing Seminar (hosted by Tom Gur) on June 19th.
- ◇ **“Rethinking Quantum Advantage via the Quantum Singular Value Transform”**
Lightning talk (5 min) at the Foundations of Quantum Computing (FQC) Conference in Edinburgh, Scotland on June 17th.
- ◇ **“A Quadratic Speedup for Finding Nash Equilibria of Quantum Zero-Sum Games”** [[Video](#)]
Invited talk (45 min) at the Surfing the Ocean ERC Seminar (virtual) on March 13th.
- ◇ **“On the Pauli Spectrum of QAC⁰”**
Invited talk (1 hour) at the UC Berkeley SAIL Seminar (hosted by Michael Jordan) on March 10th.
- ◇ **“Methods for Reducing Ancilla Overhead in Block Encodings”**
Invited talk (1 hour) at the Simons Institute Quantum Pod Seminar (hosted by Umesh Vazirani) at UC Berkeley on March 6th.
- 2024** ◇ **“Methods for Reducing Ancilla Overhead in Block Encodings”** [[Video](#)]
Contributed Talk (15 min) at the 2024 Quantum Techniques in Machine Learning (QTML) Conference in Melbourne, Australia on November 29th.
- ◇ **“Learning shallow quantum circuits with many-qubit gates”** [[Video](#)]
Contributed Talk (15 min) at the 2024 Quantum Techniques in Machine Learning (QTML) Conference in Melbourne, Australia on November 26th.
- ◇ **“Learning shallow quantum circuits with many-qubit gates”**
Invited talk (1 hour chalkboard talk) at the NYU Center for Quantum Phenomenon on October 28th.
- ◇ **“On the Pauli Spectrum of QAC⁰”** [[Video](#)]
Invited talk (1 hour) at the CWI QuSoft Seminar (virtual) on April 12th.
- ◇ **“On the Pauli Spectrum of QAC⁰”**
Invited talk (1 hour) at the UC Berkeley Theory Lunch Seminar (hosted by Avishay Tal) on March 13th.
- 2023** ◇ **“A Quadratic Speedup for Finding Nash Equilibria of Quantum Zero-Sum Games.”** [[Slides](#)]
Long talk (~6% acceptance) (30 min) at the 2023 Quantum Techniques in Machine Learning (QTML) Conference in CERN, Switzerland on November 21st.
- ◇ **“On the Pauli Spectrum of QAC⁰”**
Invited talk (3 hours) at Simons Institute Quantum Pod Seminar (hosted by Umesh Vazirani) at UC Berkeley on October 31st.
- 2019** ◇ **“Uncertainty Quantification in End-to-End Implicit Neural Representations for Medical Imaging.”** [[Video](#)]
Oral (~6% acceptance) (30 min) at the 2021 MedNeurIPS Workshop (Virtual) on December 14th.

2016 ♦ “Person-following UAVs.” [[Video](#)]

Contributed Talk (5 min) at the 2016 IEEE Winter Applications in Computer Vision (WACV) Conference at Lake Placid, New York on March 10th.

AWARDS & HONORS

- 2024** Played for the Portuguese national lacrosse team in the 2024 Women’s European Lacrosse Championships
- 2022** Oxford University Blues Award “Highest honour granted to individual sportspeople at the University of Oxford.”
- 2021** OneQuantum Leading Female Women Quantum Community Creator of the Year Award (Sponsored by QuEra)
Oxford University Half-Blue Athletic Award
- 2020** Ford Foundation PhD Fellowship Honorable Mention
- 2019** Paul and Daisy Soros Fellowship Finalist - *forfeited application upon receiving Rhodes*
Hertz PhD Fellowship Regional Finalist
MIT Churchill Nominee (1 of 2 possible MIT nominees) - *forfeited application upon receiving Rhodes*
- 2018** MIT EECS SuperUROP Draper Laboratory Undergraduate Research and Innovation Scholar
MIT School of Engineering Barry Goldwater Nominee (2 nominees from MIT each year)
MIT Society of Women Engineers Outstanding Board Member
- 2017** Minor Planet 33680 (Main-belt Asteroid, discovered 5/13/99 by Lincoln Labs) named “Vasconcelos”
- 2016** Intel International Science & Engineering Fair “Robotics & Intelligent Machines” 2nd Place Grand Award
Intel International Science & Engineering Fair WebValley Special Award
Intel Excellence in Computer Science Award
Greater San Diego Science & Engineering Fair ISEF Sweepstakes Winner (Top 6)
Southern California Junior Science and Humanities Symposium 1st Place
Scholastic Gold Key & 3 Honorable Mentions
- 2015** Intel International Science & Engineering Fair “Robotics & Intelligent Machines” 4th Place Grand Award
Intel International Science & Engineering Fair CERN Special Award
Intel International Science & Engineering Fair United Technologies Special Awards
Intel Excellence in Computer Science Award
Greater San Diego Science & Engineering Fair ISEF Sweepstakes Winner (Top 6)
2 Scholastic Silver Keys & 2 Honorable Mentions
- 2014** California Senate Award
National Honor Society (NHS) & National Art Honor Society (NAHS) Inductee
Trained with Portuguese WU17 National Soccer Team

SCHOLARSHIPS & FELLOWSHIPS

- 2024** Paul and Daisy Soros Fellowship for New Americans (\$90,000) - [1st Portuguese recipient of the fellowship.](#)
- 2019** NSF Graduate Research Fellowship
Rhodes Scholarship*
Marshall Scholarship* [Declined for Rhodes]
[*3rd student in MIT history to receive both the Rhodes and Marshall. \[Reference: Kimberly Benard\]](#)
- 2017-18** Johnson & Johnson MIT Summer Research Fellowship (\$5,520)
D.E. Shaw Latitude Fellowship (\$1,500)
Palantir Women in Technology Scholarship (\$7,000)
SWE GE Women's Network Scholarship (x2, \$10,000)
- 2015-16** SanDisk Scholar (\$10,000)
Athena Pinnacle Scholarship (\$10,000)
San Diego AFCEA Scholarship (\$8,000)
SWE Paula Loring Simon Scholarship (\$1,250)
San Diego Society of Women Engineers ViaSat Scholarship (\$1,000)
Professional Engineers in CA Government Fellowship (\$1000)
Cabrillo Civics Scholarship (\$400)
National Space Club Foundation Scholarship (\$1,000)
ISEF+JSHS Awards (\$7,000)

NEWS

- 2026** PD Soros: [Q&A: UC Berkeley PhD Student Francisca Vasconcelos Looks Back At The Fellowship](#)
- 2024** Press Minho: [European Women's Lacrosse Championship already moves in Braga](#)
Braga TV: [Braga hosts European Women's Lacrosse Championship](#)
Diario do Minho: [Portugal in group of Norway, Czechia, Austria & Ireland at Lacrosse European Championship](#)
Radio Vale do Minho: [Women's National Lacrosse Team passed through Valença](#)
MIT News: [Two from MIT awarded 2024 Paul and Daisy Soros Fellowships for New Americans](#)
Berkeley CDSS: [Francisca Vasconcelos chosen as Paul and Daisy Soros New American fellow](#)
PD Soros: [Meet the Class of 2024 Paul & Daisy Soros Fellows](#)
PD Soros: [Paul & Daisy Soros Fellowships Announce 2024 Class of Distinguished New Americans](#)
- 2022** Cherwell: [Oxford victory at 36th Women's Varsity Football match](#)
- 2020** IBM Research: [IBM and The Coding School to offer free online quantum computing course for 5,000 students](#)
MIT News: [MIT researchers lead high school educational initiative on quantum computing](#)
MIT News: [Generating photons for communication in a quantum computing system](#)
MIT News: [Cosmic rays may soon stymie quantum computing](#)
MIT News: [MIT chapter of the Phi Beta Kappa Society inducts 115 students from the Class of 2020](#)
Torrey Pines High School Falconer: [TPHS alumna awarded the Rhodes Scholarship](#)
- 2019** San Diego Union Tribune: [San Diego engineering whiz chosen to be a Rhodes Scholar](#)
MIT.edu Spotlight: [Five named Rhodes Scholars](#)
MIT News: [Five MIT students named 2020 Rhodes Scholars](#)
The Tech: [Four more MIT seniors declared 2020 Rhodes Scholars](#)
- 2017** MIT News: [Adding hands-on practice to science and engineering classes](#)
- 2016** Congressional App Challenge: [#TechTeen – Francisca Vasconcelos](#)
Congressman Scott Peters: [Quantum App Recognized](#)
San Diego Union Tribune: [2016 San Diego Science & Engineering Fair Winners](#)
Society for Science & the Public: [Society Alumni offer advice](#)
- 2015** Society for Science & the Public: [Intel ISEF alumna creates app to explain physics](#)
Torrey Pines High School Falconer: [One in 2,616: Francisca Vasconcelos](#)
San Diego Union Tribune: [Science Fair Crowns Contest Winners](#)

RESEARCH & TECHNICAL WORK EXPERIENCE

Amazon Quantum Research Intern

May-Aug 2026

Supervisors: Fernando Brandão, Alex Dalzell, András Gilyén

- project to be published

Oxford OxCSML Masters Research

Nov 2020-May 2022

Supervisor: Yee Whye Teh

- project title: “UncertaINR: Uncertainty Quantification in End-to-End Implicit Neural Representations”
- improving reliability of CT image reconstruction, via calibrated implicit neural representations
- study and characterize the relative performance of several Bayesian deep learning approaches

Microsoft Research Quantum Intern

May-Aug 2020

Supervisor: Marcus DaSilva

- project title: “Learning the Spatial Correlation Structure in Multiqubit Measurement Error”

MIT RLE Engineering Quantum Systems SuperUROP

June 2018 - May 2020

Supervisor: Morten Kjaergaard (Postdoc) ◊ Prof. William Oliver, Terry Orlando, and Simon Gustavsson’s Lab

- expanded lab’s quantum state tomography suite (based on MLE) from 2-qubit to n-qubit systems, improved runtime by 100x
- developing novel QST approach using generative adversarial networks
- programmed an FPGA (with Megan Yamoah) to speed-up lab’s measurement of the quantum computer
- improved speed and computation resources of data analysis code for waveguide QED experiment

Rigetti Computing Junior Quantum Engineering Intern

June - August 2019

Supervisors: Peter Karalekas, Marcus DaSilva ◊ Full Stack Quantum Engineering Group

- research to improve the algorithmic performance of noisy quantum devices using ensembling techniques from machine learning

MIT Media Lab Camera Culture UROP

Sept. 2017 - Sept. 2018

Supervisor: Achuta Kadambi (Grad Student) ◊ Prof. Ramesh Raskar’s Lab

- worked on LIDAR to “see through fog,” using 2-laser interferometry
- worked on learning a general transformation to boost ImageNet classification performance of major networks (i.e. AlexNet)

NASA Jet Propulsion Laboratory (JPL) Engineering Intern

June 2017 - Aug. 2017

Supervisor: Jose Velazco (Researcher) ◊ Ground Communication Group (333K)

- worked with other interns to create “smart” monitoring system for new Deep Space Network amplifiers, consisting of thousands of ESP32s connected to central database and visualization/control webpage
- modified CAD designs of Inter-Satellite Omnidirectional Optical Communicator transmitters for LIDAR functionality

MIT CSAIL NetMIT UROP

Oct. 2016 - June 2017

Supervisors: Deepak Vashit (Grad Student) & Anubhav Jain (MEng) ◊ Prof. Dina Katabi’s Lab

- developed an API and “Smart Home” application (smart alarm clock) for the group’s research on wireless location tracking
- developed location data collection iOS application to train neural network to improve localization pattern recognition

Sidus Solutions Engineering Intern

June 2015 - Aug. 2015

San Diego Marine Technology Company

- soldered micro-controllers, programmed GUIs, CADed product parts on Solidworks, fixed broken camera systems, tested products, and laser etched logo information into products

TEACHING EXPERIENCE

TCS Qubit x Qubit Intro to Quantum Computing Year-Long Course Instructor *AY 2020/21, AY 2022/23*
The Coding School Non-Profit ◊ Co-Instructor with Amir Karamlou

- developed and instructed this first-of-a-kind global, virtual ‘Intro to Quantum Computing’ course
- 8,000+ students, representing 120+ countries, with over 50% students coming from underrepresented backgrounds
- targeted towards young students - only pre-req was high school level geometry
- 99% of students would recommend the course to a friend, 96% of students rated the course as good or excellent
- 96% of students felt more confident in their STEM skills as result of the course

MIT EECS Intro to Quantum Computing IAP Course Instructor *Jan 2019, Jan 2020*
6.s089 ◊ Co-instructor with Amir Karamlou & Megan Yamoah

- 4 week crash course on quantum computing open to MIT community, no quantum mechanics knowledge required
- specifically lectured on quantum algorithms/protocols: Quantum Key Distribution, Grover search, QFT and Shor’s Algo.
- developed QuTip problem set questions and ran tutorials on QuTip and IBM Quantum Experience
- received a 6.7/7 rating on MIT course evaluations (6/20 eligible students responded)

PROFESSIONAL SERVICE

Conference Reviewer: QIP 2023, QIP 2024, ITCS 2024, TQC 2024 (2 papers), FOCS 2024, QIP 2025 (3 papers), QCTiP 2025, FOCS 2025, QEC 2025, SODA 2026, QIP 2026 (3 papers), STOC 2026, QCTiP 2026, TQC 2026 (4 papers), CCC 2026 (2 papers), FOCS 2026 (2 papers), SODA 2027

Journal Reviewer: IEEE Transactions on Communications, Nature Communications, Quantum Journal (2 papers)

Seminar Organizer: Simons Institute Quantum Pod Seminar (Fall 2024, Spring 2025)
UC Berkeley EECS SAIL Seminar (Spring 2025)

Departmental: UC Berkeley TCS Grad Admits Visit Organizer (Spring 2024),
UC Berkeley TCS Grad Application Reviewer (Spring 2025),
Grad Student Interview Coordinator for the Simons Institute Director Hiring (Spring 2025)

Other: MIT Distinguished Fellowships Application Reviewer (2023, 2024, 2025)
MIT Undergraduate Admission Maker Portfolio Reviewer (2020, 2021, 2022)

MIT IEEE URTC Conference Chair *Jan 2017 - Oct 2019*
2018 MIT IEEE Undergraduate Research Technology Conference

- led small group of students to organize 4th IEEE URTC (2018), a 3-day conference (approx. 300 attendees), at Stata Center
- there were 67 posters, 59 papers, and 16 lightning talks with 8 different conference tracks and 5 keynote speakers
- managed almost all logistics including sponsorship, publicity, registration, set-up, attendee housing, etc.
- served as Paper/Posters Chair for the 2017 conference, getting reviewers for all the submissions and keynote speakers
- serving as Conference Advisor for the 2019 conference, guiding the new conference chairs and board

OUTREACH

TCS Qubit x Qubit Founding Academic Program Director

June 2019 - Present

The Coding School Non-Profit

- founding member of The Coding School Qubit x Qubit initiative for K-12 quantum education
- created and co-lectured an 8,000+ student global, high-school 'Intro to Quantum Computing' course
- secured IBM quantum sponsorship for the course and am developing further teaching materials

MIT iQuISE Public Relations Manager

July 2019 - May 2020

MIT Interdisciplinary Quantum Information Science and Engineering program

- only undergrad board member of grad weekly seminars on quantum information, manage social media accounts and publicity

MIT Undergraduate Research Journal (MURJ) Copy Editor & Features Staff

Sept 2018 - Feb 2020

Volume 36 Fall 2018 & Volume 37 Spring 2019

- Author of "Quantum Computing @ MIT" - conducted interviews with Prof. Isaac Chuang, Prof. Dirk Englund, Prof. Aram Harrow, Prof. William Oliver; article published in MURJ Fall 2019 publication

MIT SWE Technology Chair

Nov 2016 - Nov 2019

MIT Society of Women Engineers

- organize workshops to teach SWE club members about tech topics (ie. Personal Websites, LaTeX, Quantum Computing, etc.)
- volunteer at STEM education events for local elementary and high-school girls

The Coding School Curriculum Developer, Teacher, & Advisory Board

June 2019 - May 2020

The Coding School Non-Profit

- revamped Python curriculum, taught high-school girl for 2-weeks, and created LaTeX template to be used for entire curriculum
- member of the TCS Young Professionals Advisory Board
- lead development of a Quantum Computing curriculum, resulting in the Qubit by Qubit initiative

MIT Associate Academic Advisor

Sept. 2017 - May 2018

Assistant to Prof. Dennis Freeman

- advised group of freshman on classes and navigating MIT, attended registration meetings with advisor

TPHS Math Tutoring

Nov 2012 - Jan 2015

After-School Math Tutoring Center (Unpaid)

- tutored students in all grade levels in high school math courses including geometry, algebra, and calculus

OUTREACH TALKS

2025 ◇ **“Starting Your STEM Adventure”**

Invited Talk (30 min) for The Coding School’s end of summer 2025 student celebration (NYC) on August 8th.

2024 ◇ **“Into the Quantum World”**

Invited Talk (1 hour) for Qubit by Qubit’s World Quantum Day 2024 (Virtual) on April 11th.

2023 ◇ **“What we’ve learned from training 20,000 students in QISE”**

Invited speaker and panelist on the Quantum Workforce Development Panel at the SPIE Photonics West Conference 2023 (San Francisco) on February 1st. Spoke on behalf of The Coding School non-profit CEO Kiera Peltz, presenting on K-12 quantum education work as Founding Academic Director and Lead Course Instructor for the Qubit x Qubit initiative.

2021 ◇ **“Lessons learned from teaching a year-long global, virtual intro to quantum computing course”** [[Program](#)]

Invited speaker at the IEEE Quantum Engineering Conference 2021 (virtual) on October 21st. Presenting on K-12 quantum education work as the Founding Academic Director and Lead Course Instructor for the Qubit x Qubit initiative.

◇ **“The Quantum Road Not Taken”** [[Video](#)]

Invited speaker at the Women in Quantum Summit IV Fireside Chat (virtual) on March 10th. Talking about my personal path into and work in quantum computing, inspiring younger females in the field.

◇ **“Educational Opportunities and Policy Landscape Roundtable”** [[Conference Website](#)]

Invited panel moderator for the Qubit x Qubit Diversity in Quantum Computing Conference (virtual) on February 27th. I discussed quantum education and policy with: Dr. Jessica Rosenberg (Associate Professor of Physics, George Mason University), Dr. Olivia Lanes (Quantum Researcher, IBM Quantum), Dr. Tomasz Durakiewicz (Program Director for Condensed Matter Physics, NSF), Dr. Kate Weber (Policy Lead for AI Research and Innovation, Google), and Dr. James Freericks (Professor of Physics, Georgetown University).

2016 ◇ **“Limitless Possibility”**

Invited talk (20 min) at the MIT Museum Girl’s Day “Secret Life of Robots” (Cambridge) on November 15th. Spoke about my high school engineering and computer science projects to inspire young girls in greater-Boston.